AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

SmartServe AI: Intelligent Food Demand Forecasting & Kitchen Optimization

Team ID: *GEN26-A0H-I137*

Team Name: *Team Dots*

Domain: *Intelligent Systems and Digital Futures*

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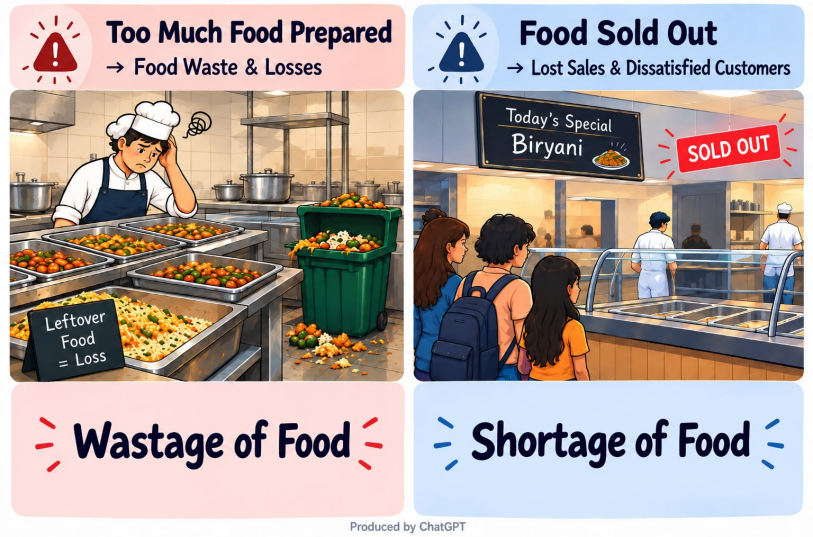
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Presentation Outline

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Problem Statement

Food-service businesses often struggle to predict customer demand and prepare the right quantity of food at the right time.
Overproduction leads to food waste and financial losses, while underproduction causes shortages, missed sales, and dissatisfied customers.
Our goal: To develop SmartServe AI, a system that predicts food demand and recommends when and how much food to prepare.



Existing Gaps & Challenges

- Many food-service businesses rely on past sales and manual estimates instead of accurate demand prediction.
- Customer demand changes with meal timings, weekdays, special occasions, and customer flow.
- Kitchen staff often lack timely alerts about when to prepare more food, increasing the risk of food waste, shortages, and financial losses.

Our Proposed Solution

- **Demand Prediction:** Uses past sales and current demand patterns to estimate how much food customers are likely to order.
- **Smart Kitchen Alerts:** Notifies kitchen staff when prepared food may run low, considering sales speed and preparation time.
- **Waste Reduction:** Recommends suitable preparation quantities to avoid unnecessary cooking and excess leftover food.
- **Staff in Control:** Gives recommendations to kitchen staff, who can review and approve preparation decisions.

Technical Approach

1. Data Collection

Collect historical sales, current orders, time of day, day of the week, and remaining prepared portions.

2. AI-Based Prediction

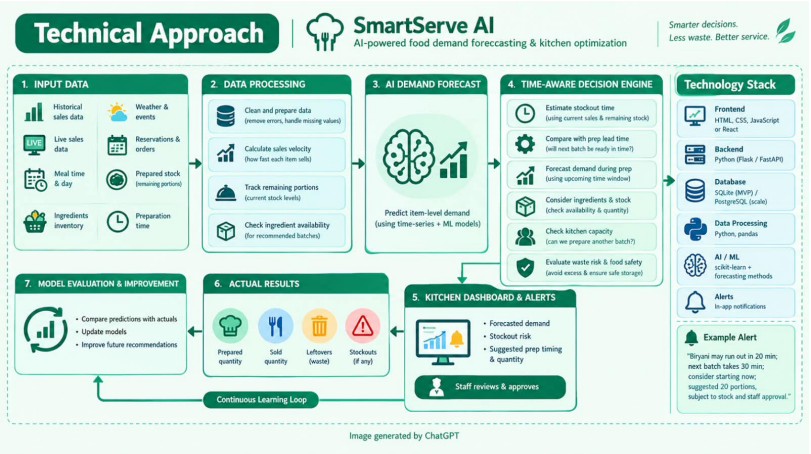
Python, Pandas, and Scikit-learn analyse demand patterns and estimate the quantity of food likely to be ordered.

3. Smart Kitchen Alerts

Compare predicted demand with available portions and preparation time to recommend when and how much to cook.

4. Staff Dashboard

Display predictions and alerts so kitchen staff can review and approve preparation decisions.



Technology Stack

Frontend: HTML, CSS, JavaScript
Backend: Python Flask
AI/ML: Pandas, Scikit-learn
Database: SQLite

Innovation & Uniqueness

- **Time-Aware Demand Prediction:** Unlike basic planning based only on past sales, SmartServe AI considers meal timings, weekdays, historical sales, and current orders to estimate changing customer demand.
- **Predictive Kitchen Alerts:** The system considers sales speed, remaining prepared portions, and food preparation time to alert kitchen staff when more food may need to be prepared before existing stock runs out.
- **Balanced Food Preparation:** Instead of encouraging kitchens to cook more food unnecessarily, SmartServe AI recommends preparation quantities based on expected demand and available food, helping reduce waste while avoiding shortages.
- **Practical Decision Support:** A simple dashboard presents demand predictions and preparation recommendations, allowing staff to review and approve decisions while considering ingredient availability and kitchen capacity.
- **Adaptable Across Food Businesses:** The approach can be adapted for

Feasibility & Scalability

- **Technical Feasibility:** Built using accessible tools such as HTML, CSS, JavaScript, Python Flask, SQLite, Pandas, and Scikit-learn for the dashboard, data storage, and demand prediction.
- **Operational Feasibility:** Uses sales records, current orders, meal timings, and available food portions. Staff can review recommendations before deciding how much food to prepare.
- **Financial Feasibility:** Open-source software and existing computer resources can help reduce the initial cost of developing the prototype.
- **Scalability:** The system can start with one outlet and later expand to restaurants, hotels, cafés, canteens, and cloud kitchens.
- **Future Expansion:** Possible upgrades include inventory tracking, weather-based predictions, multiple-outlet management, and integration with existing billing systems.

Cost for Product (Optional)

- **Development Tools:** HTML, CSS, JavaScript, Python Flask, SQLite, Pandas, and Scikit-learn. These tools are open-source and can be used without software licence fees.
- **Development and Testing:** Estimated cost: 1,000–3,000 for possible testing, internet, and miscellaneous development expenses.
- **Hosting and Deployment:** Estimated cost: 1,000–3,000 for basic hosting and deployment, depending on the service selected.
- **Miscellaneous Expenses:** Estimated cost: 1,000–3,000 for documentation, demonstrations, and other project requirements.
- **Estimated Total: 3,000–9,000** for a basic prototype, assuming existing computers are available.

Use Cases (compulsorily 3 use cases must be there)

- **The Problem:** Restaurants and hotels experience different demand levels during breakfast, lunch, dinner, weekends, and special occasions. Preparing too much food leads to waste, while preparing too little can result in lost sales and unhappy customers.
- **How SmartServe AI Helps:** The system analyses historical sales, current orders, meal timings, and remaining prepared portions to estimate upcoming demand. It recommends preparation quantities and considers how long each dish takes to prepare.
- **Example Scenario:** Suppose a restaurant sells biryani quickly during lunch. SmartServe AI estimates how many portions may be needed and considers the preparation time before alerting staff to prepare another batch.
- **Expected Business Benefit:** Better preparation planning, fewer avoidable shortages, reduced food waste, and more efficient kitchen operations.

Use Case: College Canteens

- **The Problem:** College canteens experience sudden increases in demand during lunch breaks and between classes. Student attendance, class schedules, and daily preferences can affect how much food is sold. This makes it difficult to prepare the right quantity in advance.
- **How SmartServe AI Helps:** The system studies previous sales patterns, meal timings, current orders, and available portions to estimate demand for different dishes. It can also consider class schedules or special campus events when this information is available.
- **Example Scenario:** Suppose a canteen usually sells around 100 portions of biryani during lunch. On a day with a campus event, demand may increase. If sales are rising faster than expected, SmartServe AI can alert staff to review whether another batch should be prepared, considering preparation time and available ingredients.
- **Expected Business Benefit:** Better lunch-hour planning, fewer students facing unexpected shortages, less leftover food, and improved use of ingredients and kitchen resources.

Use Case: Cloud Kitchens & Food Chains

- **The Problem:** Cloud kitchens and food chains receive orders through different channels. Demand can change rapidly during peak hours, weekends, and promotional offers. Poor coordination may lead to delayed orders, unnecessary preparation, or popular dishes becoming unavailable.
- **How SmartServe AI Helps:** Using available order and sales data, the system estimates demand for different dishes and compares it with remaining prepared portions. It considers sales speed, preparation time, and ingredient availability to recommend when additional preparation may be needed.
- **Example Scenario:** Suppose online orders for a particular meal increase sharply during dinner. SmartServe AI detects the rise in demand and alerts kitchen staff before the available portions are likely to run out, giving them time to review and prepare another batch if required.
- **Expected Business Benefit:** Improved order planning, better kitchen coordination, fewer avoidable stockouts, and more controlled food preparation across outlets.

Impacts and Benefits

- **Environmental Impact:** By helping kitchens prepare food according to predicted demand, SmartServe AI aims to reduce avoidable food waste and the resources wasted in producing excess food.
- **Financial Benefits:** Better demand estimation can help businesses control ingredient costs, reduce unnecessary preparation, and avoid losing sales when popular dishes become unavailable.
- **Operational Efficiency:** Timely preparation alerts can help kitchen staff plan their work, manage busy periods, and coordinate food preparation more effectively.
- **Customer Satisfaction:** Improved food availability can reduce unexpected shortages and help customers receive their preferred dishes more consistently.
- **Social Impact:** More efficient food planning can encourage responsible food management and support a culture of reducing food waste across different food-service businesses.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 12	12.3	Responsible Consumption and Production	SmartServe AI aims to reduce food waste by predicting demand and recommending suitable preparation quantities. This supports more responsible food consumption and production.
SDG 9	9.5	Industry, Innovation and Infrastructure	The project applies AI-based demand forecasting and digital tools to improve food-service operations and encourage technology-driven innovation.

Individual Member Contribution

Sneha Sahu	Team leadership, problem identification, project planning, and coordination of the SmartServe AI solution.
K.V Rubika	Research on existing food-demand management approaches, problem analysis, and documentation.
Katta Hamshika Padmini	Research into demand prediction, proposed AI/ML approach, and technical workflow planning.
Chittem Amrutha Varshini	Use-case development, expected impact analysis, and presentation preparation.

Yukti ID

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